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PROG6212_POE_PART_1
ST10492744

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Section A – ERD

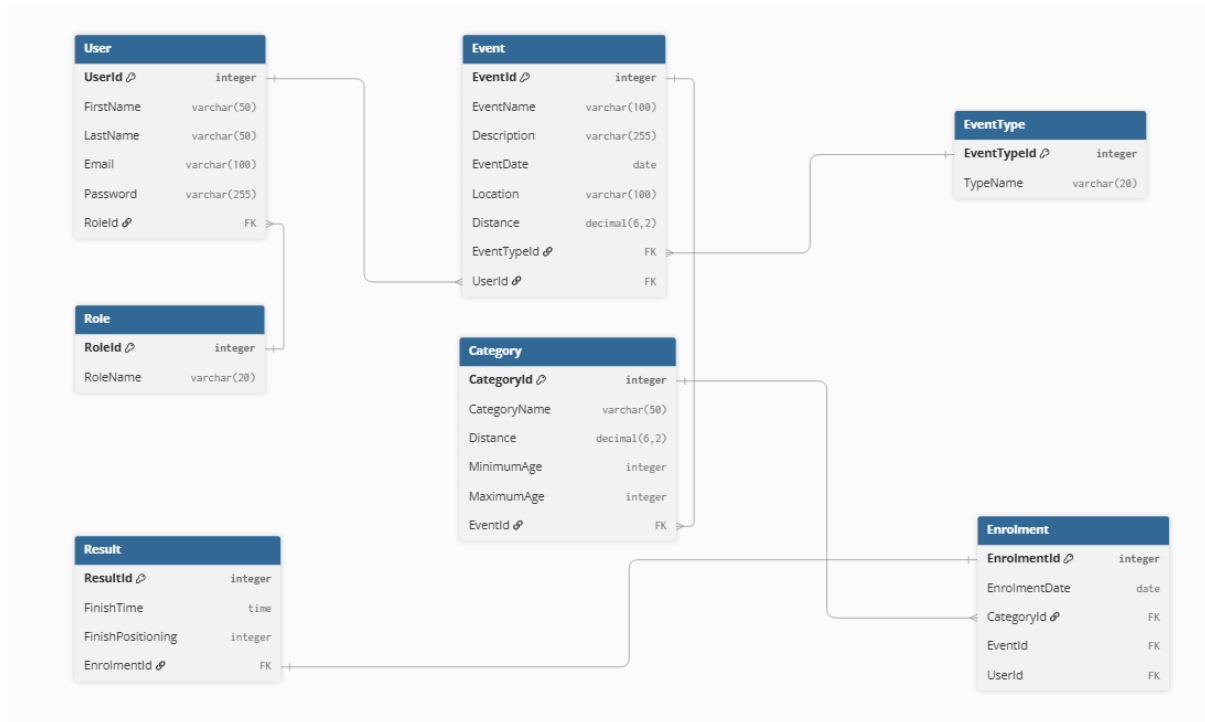


Figure 1 ERD

```

1  Table User {
2      UserId integer PK
3      FirstName varchar(50)
4      LastName varchar(50)
5      Email varchar(100)
6      Password varchar(255)
7      RoleId FK
8  }
9
10 Table Role {
11     RoleId integer PK
12     RoleName varchar(20)
13 }
14
15 Table Event {
16     EventId integer PK
17     EventName varchar(100)
18     Description varchar(255)
19     EventDate date
20     Location varchar(100)
21     Distance decimal(6,2)
22     EventTypeId FK
23     UserId FK
24 }
25
26 Table EventType{
27     EventTypeId integer PK
28     TypeName varchar(20)
29 }
30
31 Table Category {
32     CategoryId integer PK
33     CategoryName varchar(50)
34     Distance decimal(6,2)
35     MinimumAge integer
36     MaximumAge integer
37     EventId FK
38 }

```

Figure 2 ERD_code_1

```

40 Table Enrolment {
41     EnrolmentId integer PK
42     EnrolmentDate date
43     CategoryId FK
44     EventId FK
45     UserId FK
46 }
47
48 Table Result {
49     ResultId integer PK
50     FinishTime time
51     FinishPositioning integer
52     EnrolmentId FK
53 }
54
55 //One-to-many
56 Ref: Role.RoleId < User.RoleId
57
58 //One-to-many
59 Ref: User.UserId < Event.UserId
60
61 //One-to-many
62 Ref: Event.EventId < Category.EventId
63
64 //One-to-many
65 Ref: Category.CategoryId < Enrolment.CategoryId
66
67 //One-to-many
68 Ref: EventType.EventTypeid < Event.EventTypeid
69
70 //One-to-one
71 Ref: Enrolment.EnrolmentId - Result.EnrolmentId

```

Figure 3 ERD_code_2

Section B – API Endpoint plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user	Public	FirstName, LastName, Email, Password, RoleID	User registered successfully
POST	/api/auth/login	Log in an existing user	Public	Email, Password	Authentication result and user details
GET	/api/users/profile	View the logged-in user's profile	Organiser / Participant	None	User profile
PUT	/api/users/profile	Update the logged-in user's profile	Organiser / Participant	Profile information	Updated profile
GET	/api/events	View all available events	Organiser / Participant	None	List of events
GET	/api/events/{id}	View details of a specific event	Organiser / Participant	None	Event details
POST	/api/events	Create a new event	Organiser	EventName, Description, EventDate, Location, Distance, EventTypeID	Created event

PUT	/api/events/{id}	Update an event	Organiser	Event details	Updated event
DELETE	/api/events/{id}	Delete an event	Organiser	None	Confirmation message
GET	/api/events/{id}/categories	View categories for an event	Organiser / Participant	None	List of categories
POST	/api/events/{id}/categories	Create a category for an event	Organiser	CategoryName, MinimumAge, MaximumAge, Distance	Created category
PUT	/api/categories/{id}	Update an event category	Organiser	Category information	Updated category
DELETE	/api/categories/{id}	Delete an event category	Organiser	None	Confirmation message
POST	/api/enrolments	Enrol a participant in an event and category	Participant	EventID, CategoryID	Enrolment created
GET	/api/events/{id}/enrolments	View enrolments for an event	Organiser	None	List of enrolments
GET	/api/results/me	View the participant's own results	Participant	None	List of own results

POST	/api/results	Record a participant's result	Organiser	EnrolmentID, Finish Time, Finishing Position	Result created
GET	/api/events/{id}/results	View results for an event	Organiser	None	List of event results

Section C – SQL Script

Viewing data

```

220  -- View all users
221  SELECT * FROM [User];
222
223  -- View all events
224  SELECT * FROM Event;
225
226  -- View all event types
227  SELECT * FROM EventType;
228
229  -- View all categories
230  SELECT * FROM Category;
231
232  -- View all enrolments
233  SELECT * FROM Enrolment;
234
235  -- View all results
236  SELECT * FROM Result;
237
238  -- View all roles
239  SELECT * FROM Role;
240

```

100 % No issues found

Results Messages

	UserID	FirstName	LastName	Email	Password	RoleID
1	1	John	Mokoena	john@raceday.com	Password123	1
2	2	Sarah	Williams	sarah@raceday.com	Password456	1
3	3	Thabo	Dlamini	thabo@email.com	Password789	2
4	4	Lerato	Molefe	lerato@email.com	Password321	2

```

223 -- View all events
224 SELECT * FROM Event;
225
226 -- View all event types
227 SELECT * FROM EventType;
228
229 -- View all categories
230 SELECT * FROM Category;
231
232 -- View all enrolments
233 SELECT * FROM Enrolment;
234
235 -- View all results
236 SELECT * FROM Result;
237
238 -- View all roles
239 SELECT * FROM Role;
240

```

100 % No issues found

Results

Messages

	EventID	EventName	Description	EventDate	Location	Distance	EventTypeID	UserID
1	1	Johannesburg City Run	Annual city running event	2026-10-10	Johannesburg	10.00	1	1
2	2	Soweto Community Walk	Community health and fitness walk	2026-10-17	Soweto	5.00	2	2
3	3	Pretoria Cycle Challenge	Annual cycling challenge	2026-11-01	Pretoria	21.00	3	1

```

226 -- View all event types
227 SELECT * FROM EventType;
228
229 -- View all categories
230 SELECT * FROM Category;
231
232 -- View all enrolments
233 SELECT * FROM Enrolment;
234
235 -- View all results
236 SELECT * FROM Result;
237
238 -- View all roles
239 SELECT * FROM Role;
240

```

100 % No issues found

Results		Messages	
	EventTypeID	TypeName	
1	3	Cycle	
2	1	Run	
3	2	Walk	


```

229  -- View all categories
230  SELECT * FROM Category;
231
232  -- View all enrolments
233  SELECT * FROM Enrolment;
234
235  -- View all results
236  SELECT * FROM Result;
237
238  -- View all roles
239  SELECT * FROM Role;
240

```

100 % No issues found

Results

Messages

	CategoryID	CategoryName	Distance	MinimumAge	MaximumAge	EventID
1	1	Under 20	10.00	13	19	1
2	2	Senior	10.00	20	59	1
3	3	Veteran	10.00	60	NULL	1
4	4	Under 20	5.00	13	19	2
5	5	Senior	5.00	20	59	2
6	6	Veteran	5.00	60	NULL	2
7	7	10km Cycle	10.00	16	59	3
8	8	21km Cycle	21.00	16	59	3
9	9	Veteran Cycle	21.00	60	NULL	3

```

232  -- View all enrolments
233  SELECT * FROM Enrolment;
234
235  -- View all results
236  SELECT * FROM Result;
237
238  -- View all roles
239  SELECT * FROM Role;
240

```

100 % No issues found

Results					
	EnrolmentID	EnrolmentDate	CategoryID	EventID	UserID
1	1	2026-09-01	1	1	3
2	2	2026-09-02	2	1	4
3	3	2026-09-03	4	2	3
4	4	2026-09-03	5	2	4
5	5	2026-09-04	8	3	3

```

235  -- View all results
236  SELECT * FROM Result;
237
238  -- View all roles
239  SELECT * FROM Role;
240

```

100 % No issues found

Results				
	ResultID	FinishTime	FinishPosition	EnrolmentID
1	1	01:02:35.0000000	15	1
2	2	01:08:20.0000000	28	2

```
238 -- View all roles
239 SELECT * FROM Role;
240
```

100 % No issues found

Results		Messages
	RoleID	RoleName
1	1	Organiser
2	2	Participant